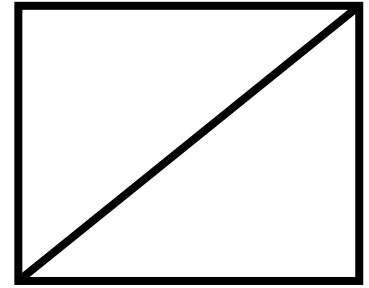


SULIT



**KEMENTERIAN PENDIDIKAN
Bahagian Matrikulasi**

**PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI (PSPM)
SEMESTER II & IV, SESI 2022/2023**

**ELECTRICAL & ELECTRONICS ENGINEERING 2
EE025**

SKEMA JAWAPAN & PEMARKAHAN

PEMERIKSA	
KETUA KUMPULAN PEMERIKSA	
KETUA PEMERIKSA	

Catatan:

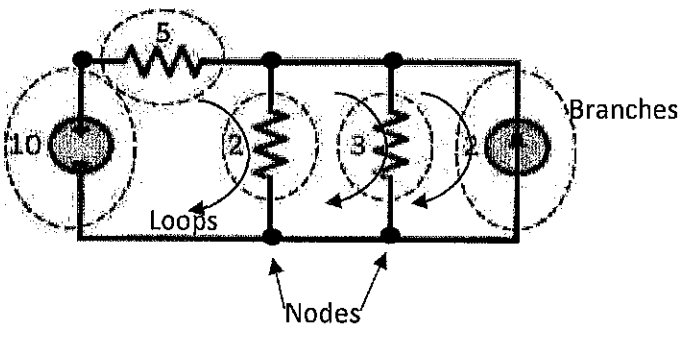
Dokumen ini wajib dikembalikan selepas tamat bengkel kerja pemeriksaan.

SULIT

ANSWER SCHEME: ELECTRICAL AND ELECTRONIC ENGINEERING 2

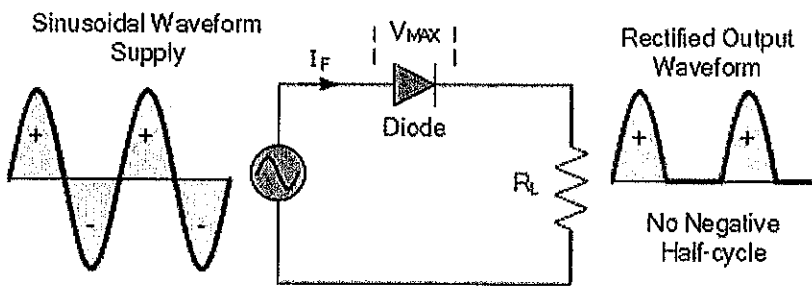
CODE COURSE: EE025

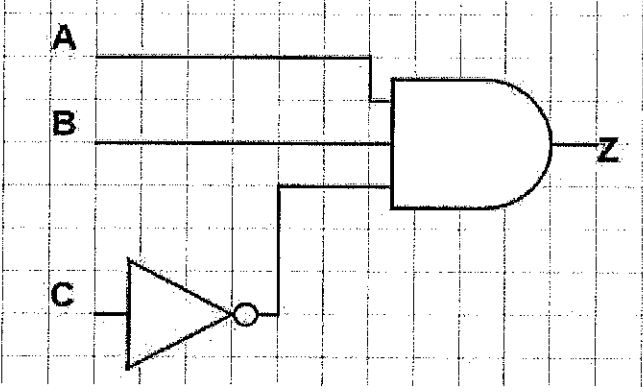
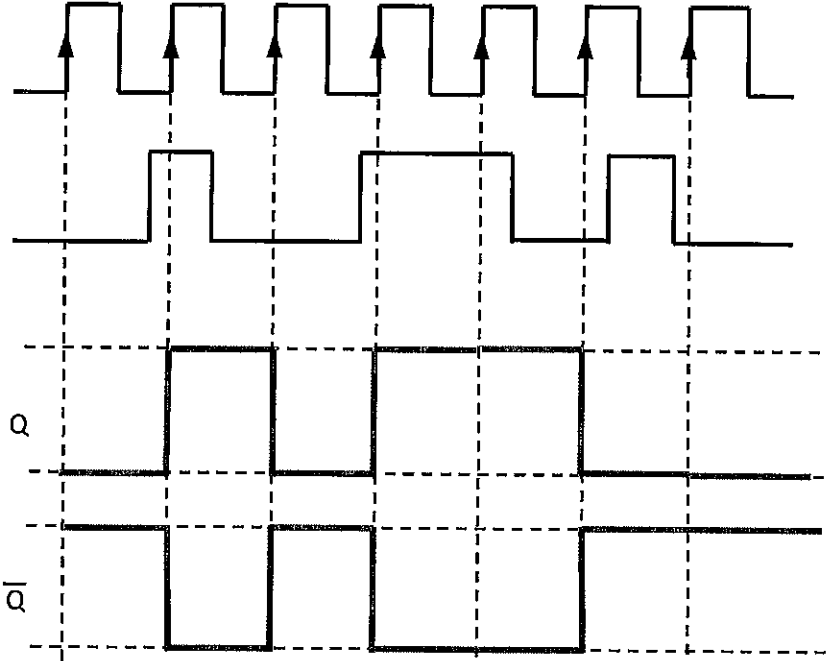
Set 1

NO	PART	SCHEME	MARKS
1	a)	Node: A point where two or more circuit elements' terminals are connected together Branch: The portion of the circuit between two nodes which can deliver or absorb energy Loop: Formed from a node passing through a set of nodes and returning to the starting point or node without passing the same node twice or more. 	1 1 1 2
	b) (i)	Loop 1: $22i_1 - 12i_2 - 3i_3 = 0$ Loop 2: $-12i_1 + 24i_2 - 12i_3 = 63$ Loop 3: $-3i_1 - 12i_2 + 26i_3 = 0$	2 2 2
	(ii)	By using calculator, simultaneous equation: $i_x = i_1 = 3.82 \text{ A}$	2
	(iii)	From simultaneous equation: $i_2 = 6.18 \text{ A}$ $i_3 = 3.29 \text{ A}$ $V_{R3\Omega} = (i_1 - i_3)(3) = 1.59 \text{ V}$	1 1
		TOTAL	15

2	a)	(i) From the waveform, One cycle duty (Period), $T = 50 \mu s$ Frequency, $f = \frac{1}{T} = \frac{1}{50 \mu s} = 20 \text{ kHz}$ (ii) Phase different, θ between i_1 and i_2, $\theta = \frac{t_1 - t_o}{T} \times 360^\circ = \frac{18.1 \mu - 25 \mu}{50 \mu} \times 360^\circ = -50^\circ$ (iii) Equation for current $i_{1(t)}$ $i_{p1} = 60 \text{ mA}$ $i_1(t) = i_{p1} \sin(2\pi ft - 0^\circ)$ $i_1(t) = 60 \sin(2\pi(20 \times 10^3)t - 0^\circ) \text{ mA}$ $i_1(t) = 60 \sin(40000\pi t - 0^\circ) \text{ mA}$ Equation for current $i_{2(t)}$ $i_{p2} = 80 \text{ mA}$ $i_2(t) = i_{p2} \sin(2\pi ft - \theta)$ $i_2(t) = 80 \sin(2\pi(20 \times 10^3)t - (-50^\circ)) \text{ mA}$ $i_2(t) = 80 \sin(40000\pi t + 50^\circ) \text{ mA}$ (b) (i) Vphasor (same phase with I_R): $V = I_R(R) = (0.255 \angle 0^\circ) (470 \angle 0^\circ) = 120 \angle 0^\circ \text{ V}$ (ii) R and L in parallel, $I_L = \frac{V_L}{jX_L} = \frac{120 \angle 0^\circ}{2\pi fL \angle 90^\circ} = \frac{120 \angle 0^\circ}{2\pi(400)(250 \times 10^{-3}) \angle 90^\circ}$ $= \frac{120 \angle 0^\circ}{628.32 \angle 90^\circ} = 191 \angle -90^\circ \text{ mA}$ By KCL: $I = I_R + I_L = 255 - j191 \text{ mA}$ $= 318.6 \angle -36.83^\circ \text{ mA}$	1 1 2 1 2 1 2 1 2 2 15
		TOTAL	15

3	(a)	<u>Relative permeability</u> -the ratio between magnetic flux density within a material and magnetic flux density in vacuum produced by the same strength of magnetic field. -This is similar if the permeability of the material μ is compared to the permeability of the free space μ_0. <u>Faraday law:</u> Faraday's law of electromagnetic induction states that the electromotive force (EMF) induced in a closed loop is proportional to the rate of change of magnetic flux through the loop. In mathematical terms, $EMF = -d\Phi/dt$, where Φ is the magnetic flux and t is time. This law is a fundamental concept in electrical engineering and is the basis for the operation of many devices, such as transformers, generators, and inductors.	2
	(b) (i)	Reluctance for part A: $\mathfrak{R}_A = \frac{l_1}{\mu_1 A_1} = \frac{30 \times 10^{-2}}{(1.26 \times 10^{-4})(14 \times 10^{-4})} = 1.7 \text{ M At/Wb}$	2
	(ii)	Reluctance for part B: $\mathfrak{R}_B = \frac{l_2}{\mu_2 A_2} = \frac{10 \times 10^{-2}}{(6.3 \times 10^{-3})(26 \times 10^{-4})} = 6.11 \text{ k At/Wb}$	2
	(iii)	Total of reluctance: $\mathfrak{R}_A + \mathfrak{R}_B = (1.7 \times 10^6) + (6.11 \times 10^3) = 1.706 \text{ M At/Wb}$	2
		TOTAL	10
4	a) (i)	N-type and P-type	2
	(ii)	N-type: Forms by adding pentavalent impurity atoms to the intrinsic semiconductor called donor atoms Examples: arsenic, phosphorus, antimony P-type: Forms by adding trivalent impurity atoms to the intrinsic semiconductor and these impurities are called acceptors Examples: aluminium, boron, gallium	3
			3

	(b)	 1. Correct arrangement of diode polarity - 2 marks 2. Correct flow of forward current and reverse current – 1 mark 3. Correct AC source and load – 1 mark	3 4														
	(c)	(i) Type: NPN transistor – Common Emitter configuration (ii) $V_E = I_E R_E = I_C R_E = (2 \times 10^{-3})(1 \times 10^3) = 2\text{ V}$ $V_B = V_{BE} + V_E = 0.7 + 2 = 2.7\text{ V}$ (iii) Using voltage divider method $V_B = 2.7\text{ V} = \frac{R_2}{R_1 + R_2} \times V_{CC}$ $\therefore R_1 = \left(\frac{R_2}{V_B} \times V_{CC} \right) - R_2 = \left(\frac{22k}{2.7} \times 16 \right) - 22k = 108.4\text{ k}\Omega$	2 2 2 1 3														
		TOTAL	25														
5	a)	A E 6															
	(i)	1010 1110 0110	2														
		$\therefore AE_{16} = 1010\ 1110\ 0110_2$															
	(ii)	<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>16</td> <td>864</td> <td>remainder</td> <td></td> </tr> <tr> <td>16</td> <td>54</td> <td>0</td> <td rowspan="3" style="text-align: center; vertical-align: middle;">↑ (LSB)</td> </tr> <tr> <td>16</td> <td>3</td> <td>6</td> </tr> <tr> <td></td> <td>1</td> <td>3</td> </tr> </table> (MSB) $\therefore 864_{10} = 360_{16}$	16	864	remainder		16	54	0	↑ (LSB)	16	3	6		1	3	2
16	864	remainder															
16	54	0	↑ (LSB)														
16	3	6															
	1	3															

	(iii)		2
		TOTAL	20
6	(a)		5
	(b) (i)	Inputs: Temperature sensor, Salinity sensor, Ammonia sensor Outputs: green light, red light, siren, water pump	4

	(ii)	Temperature Sensor Salinity Sensor Ammonia Sensor CR1 CR1 Green Light CR1 Red Light Siren Water Pump	6
		TOTAL	15
		TOTAL	100